4-23-26 (WEEK 9)

Shun/修海 (@shun4midx)

EXAMPLE CONTINUED

Let S=signal, X=noise, $S+X$=received signal

By Law of Total Probability, $\mathbb{P}(\text{error}) = \mathbb{P}(\text{error}\mid S=0)\mathbb{P}(S=0) + \mathbb{P}(\text{error}\mid S=1)\mathbb{P}(S=1)$

$$= \mathbb{P}(S+X \geq 0.5 \mid S=0)\mathbb{P}(S=0) + \mathbb{P}(S+X<0.5\mid S=1)\mathbb{P}(S=1)$$

$$= \mathbb{P}(X\geq 0.5)\mathbb{P}(S=0) + \mathbb{P}(X<-0.5)\mathbb{P}(S=1)$$

By symmetry, $\mathbb{P}(X\geq 0.5) = \mathbb{P}(X<-0.5)$, so $\mathbb{P}(\text{error}) = \mathbb{P}(X<-0.5)\mathbb{P}(S=0) + \mathbb{P}(X<-0.5)\mathbb{P}(S=1) = \mathbb{P}(X<-0.5)$ ($\mathbb{P}(S=0)+\mathbb{P}(S=1)=1$)

Now, we standardize. $Z = \frac{X-0}{0.3} \Rightarrow \mathbb{P}(\text{error}) = \mathbb{P}(Z \leq \frac{-0.5-0}{0.3} \approx -1.67) = \mathbb{P}(Z\geq 1.67) = 1-\mathbb{P}(Z\leq 1.67) = 1-\Phi(1.67) \approx 1-0.9525 = 0.0475$

APPROXIMATING BINOMIALS

If $np(1-p) \longrightarrow \infty$, $\text{Bin}(n,p)$ "looks normal"

THEOREM (DE-MOIVRE-LAPLACE)

Let $S_n \sim \text{Bin}(n,p)$, $p\in(0,1)$: fixed.

For fixed $a<b$, $\mathbb{P}\left(a \leq \frac{S_n-np}{\sqrt{np(1-p)}} \leq b\right) \xrightarrow{n\to\infty} \Phi(b)-\Phi(a)$, i.e. $\frac{S_n-np}{\sqrt{np(1-p)}}$ behaves like $N(0,1) \Rightarrow S_n$ behaves like $N(np, np(1-p))$

REMARK

This approx works well for $np(1-p)\geq 10$ ☺

CONTINUITY CORRECTION

If $S_n \sim \text{Bin}(n,p)$ and we estimate it with a normal distribution, then we should correct for the continuity: $\{S_n=k\} \Leftrightarrow \{k-\frac{1}{2}<S_n<k+\frac{1}{2}\}$

EXAMPLE

Toss a fair coin 40 times.

What's the probability of seeing 20 heads?

Solution

$\#\text{heads} =: S_n \sim \text{Bin}(40, \frac{1}{2})$

$\mathbb{P}(S_n=20) = \binom{40}{20}(\frac{1}{2})^{20}(\frac{1}{2})^{40-20} = \binom{40}{20}(\frac{1}{2})^{40} \approx 0.1254$

Normal approximation: $\mathbb{P}(S_n=20) \overset{\text{continuity correction}}{=} \mathbb{P}(19.5\leq S_n\leq 20.5) \overset{\text{standardize}}{=} \mathbb{P}\left(-\frac{0.5}{\sqrt{10}} \leq \frac{S_n-20}{\sqrt{10}} \leq \frac{0.5}{\sqrt{10}}\right) \overset{\text{Thm } (\because np(1-p)=10)}{\approx} \Phi\left(\frac{0.5}{\sqrt{10}}\right) - \Phi\left(-\frac{0.5}{\sqrt{10}}\right) = \Phi\left(\frac{0.5}{\sqrt{10}}\right) - \left[1-\Phi\left(\frac{0.5}{\sqrt{10}}\right)\right] = 2\Phi\left(\frac{0.5}{\sqrt{10}}\right)-1 \approx 0.1272$

EXAMPLE (POLLING)

Country has population N

Gov wants to do something but

↳ First need to measure public support

Suppose $p\%$ of the population are in support

Government polls n people w/ replacement unif at random

$\#\text{tve responses} = S_n \sim \text{Bin}(n,p)$ ∴ used to approx p

Q: What should n be to estimate p to within 1% (margin of error) with prob $\geq 95\%$ (confidence interval)?

Solution

$\{\text{Estimating } p \text{ to within } 1\%\} =: \{\text{accurate}\} = \{|\frac{S_n}{n}-p|\leq 0.01\} = \{-0.01n \leq S_n-np \leq 0.01n\} = \{-\frac{0.01n}{\sqrt{np(1-p)}} \leq \frac{S_n-np}{\sqrt{np(1-p)}} \leq \frac{0.01n}{\sqrt{np(1-p)}}\}$. However, notice $\frac{0.01n}{\sqrt{np(1-p)}} = \frac{0.01\sqrt{n}}{\sqrt{p(1-p)}} \geq 0.02\sqrt{n}$

$\therefore \mathbb{P}(\text{accurate}) = \mathbb{P}\left(-\frac{0.01n}{\sqrt{np(1-p)}} \leq \frac{S_n-np}{\sqrt{np(1-p)}} \leq \frac{0.01n}{\sqrt{np(1-p)}}\right) \geq \mathbb{P}\left(-0.02\sqrt{n} \leq \frac{S_n-np}{\sqrt{np(1-p)}} \leq 0.02\sqrt{n}\right) \approx \Phi(0.02\sqrt{n}) - \Phi(-0.02\sqrt{n}) = 2\Phi(0.02\sqrt{n})-1$

We know $\mathbb{P}(\text{accurate}) \geq 0.95$ & $\Phi(0.02\sqrt{n}) \geq 0.975 \Rightarrow 0.02\sqrt{n} \geq 1.96 \Rightarrow n \geq 9604$ (indep of population size!!)

GAMMA DISTRIBUTION

Shun/修海 (@shun4midx)

POISSON PROCESS

Rare events w/ rate $\lambda \in \mathbb{R}_{>0}$, waiting time between events $\sim \underbrace{\text{Exp}(\lambda)}_{\text{memoryless}}$

$\Rightarrow$ $\text{Exp}(\lambda)$ measures the timestamp of the first event

$\Rightarrow$ $\text{Poi}(\lambda)$ = # events occuring per unit time

$X_n :=$ timestamp of nth element

Then, $X_1 \sim \text{Exp}(\lambda)$

CDF of X_n: $F_{X_n}(t) = \mathbb{P}(X_n \leq t) = \mathbb{P}(\text{at least } n \text{ events occur in } [0,t]) = \mathbb{P}(\text{Poi}(t\lambda) \geq n) = \sum_{j=n}^{\infty} \mathbb{P}(\text{Poi}(t\lambda) = j) = \sum_{j=n}^{\infty} e^{-t\lambda} \frac{(t\lambda)^j}{j!}$

counting # events in time interval = Poisson

PDF of X_n: $f_{X_n}(t) = \frac{d}{dt}(F_{X_n}(t)) = \sum_{j=n}^{\infty} \left[\frac{d}{dt}\left(\frac{e^{-t\lambda}(t\lambda)^j}{j!}\right)\right] = \sum_{j=n}^{\infty} \frac{1}{j!}\left[-\lambda e^{-t\lambda}(t\lambda)^j + e^{-t\lambda}\lambda j (t\lambda)^{j-1}\right] = -\lambda \sum_{j=n}^{\infty} e^{-t\lambda}\frac{(t\lambda)^j}{j!} + \lambda \sum_{j=n}^{\infty} e^{-t\lambda}\frac{(t\lambda)^{j-1}}{(j-1)!} = \left(-\lambda \sum_{j=n}^{\infty} e^{-t\lambda}\frac{(t\lambda)^j}{j!}\right) + \left(\lambda \sum_{j=n-1}^{\infty} e^{-t\lambda}\frac{(t\lambda)^j}{j!}\right)$

$= \frac{\lambda e^{-t\lambda}(t\lambda)^{n-1}}{(n-1)!}$ $(t \geq 0)$

Which, notice, when $n=1$, $f_{X_1}(t) = \lambda e^{-t\lambda} \rightarrow$ exp PDF

DEFINITION

Gamma distribution w/ param $\lambda \in \mathbb{R}_{>0}$, $\alpha \in \mathbb{R}_{>0}$

We say $X \sim \Gamma(\alpha, \lambda)$

Then, $f(x) = \begin{cases} \frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)}, & x \geq 0 \\ 0, & x > 0 \end{cases}$, where $\Gamma(\alpha) := \int_0^{\infty} e^{-y} y^{\alpha-1} dy$, where $\Gamma(\alpha) = (\alpha-1)!$ $\forall \alpha \in \mathbb{N}$

$\hookrightarrow$ $\Gamma(n, \lambda)$ = timestamp for the nth event in the Poisson process w/ rate λ, $\forall n \in \mathbb{N}$

EXPECTED VALUE

$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x \cdot \frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)} dx = \frac{1}{\Gamma(\alpha)} \int_0^{\infty} e^{-\lambda x}(\lambda x)^{\alpha} dx \overset{y=\lambda x}{=} \frac{1}{\Gamma(\alpha)} \int_0^{\infty} e^{-y} y^{\alpha} \cdot \frac{1}{\lambda} dy = \frac{1}{\lambda \Gamma(\alpha)} \int_0^{\infty} e^{-y} y^{(\alpha+1)-1} dy = \frac{\Gamma(\alpha+1)}{\lambda \Gamma(\alpha)} = \frac{\alpha}{\lambda}$

SPECIAL CASE

$\alpha = n \in \mathbb{N}$, $\Gamma(n, \lambda)$ = timestamp of nth event in the Poisson process w/ $\text{Exp}(\lambda)$ waiting time

Say $X \sim \Gamma(n, \lambda)$, then $X = W_1 + W_2 + \ldots + W_n$, where W_i = waiting time btwn $(i-1)^{th}$ and i^{th} events $\sim \text{Exp}(\lambda)$

By linearity of expectation, $\mathbb{E}[X] = \mathbb{E}[\sum_{i=1}^{n} W_i] = \sum_{i=1}^{n} \mathbb{E}[W_i] = \sum_{i=1}^{n} \frac{1}{\lambda} = \frac{n}{\lambda}$

VARIANCE

$X \sim \Gamma(\alpha, \lambda)$, then $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$

$\mathbb{E}[X^2] = \int_0^{\infty} x^2 \frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)} dx = \frac{1}{\lambda \Gamma(\alpha)} \int_0^{\infty} e^{-\lambda x}(\lambda x)^{\alpha+1} dx = \frac{1}{\lambda^2 \Gamma(\alpha)} \int_0^{\infty} e^{-y} y^{\alpha+1} dy = \frac{\Gamma(\alpha+2)}{\lambda^2 \Gamma(\alpha)} = \frac{\alpha(\alpha+1)}{\lambda^2}$

$\therefore$ $\text{Var}(X) = \frac{\alpha(\alpha+1)}{\lambda^2} - \left(\frac{\alpha}{\lambda}\right)^2 = \frac{\alpha}{\lambda^2}$